TASKFLOW – MANUAL TESTING USING APACHE JMETER

Software Engineering / Testing Assignment

Project: Taskflow – Daily Task & Check-in Application

Tool: Apache JMeter 5.6.3 | Runtime: Java 25.0.2 | Platform: Windows

Student: _____ Roll/MIS: _____

Division/Batch: _____ Date: _____

1. Objective

The objective is to perform structured manual functional testing of the Taskflow application using Apache JMeter. The testing covers application availability, task creation, retrieval, update, deletion, and handling of an invalid task request. JMeter is used to configure and execute HTTP requests and observe response codes, response times, and errors.

2. Testing Environment

Component	Configuration
Operating System	Microsoft Windows
Application	Taskflow
Backend	FastAPI demonstration API
Server	127.0.0.1:8000
Testing Tool	Apache JMeter 5.6.3
Java	Java 25.0.2
Test Type	Manual functional/API testing using JMeter

3. Test Strategy

A single-thread functional test is executed first to verify correctness. The Thread Group can then be increased to simulate multiple users if a load-testing demonstration is required. The test plan uses HTTP samplers for the Taskflow REST endpoints and listeners for result inspection.

Test flow: Health Check → Open Taskflow → List Tasks → Create Task → Get Task → Update Task → Delete Task → Invalid Task.

4. Test Cases

ID	Test Case	Request	Expected	Status
TC-01	Health Check	GET /health	200 / status=healthy	To be filled after execution
TC-02	Open Taskflow	GET /	200 / Taskflow API ready	To be filled after execution
TC-03	List Tasks	GET /tasks	200 / JSON array returned	To be filled after execution
TC-04	Create Task	POST /tasks	201 / Task object with id	To be filled after execution
TC-05	Get Task	GET /tasks/{id}	200 / Created task returned	To be filled after execution
TC-06	Update Task	PUT /tasks/{id}	200 / Updated values returned	To be filled after execution
TC-07	Delete Task	DELETE /tasks/{id}	200 / Deletion confirmation	To be filled after execution
TC-08	Invalid Task	GET /tasks/999999	404 / Task not found	To be filled after execution

5. JMeter Test Plan Configuration

The supplied JMX file contains a Test Plan with a Thread Group, HTTP Request samplers, JSON headers for POST/PUT operations, a dynamic task-ID capture step, View Results Tree, and Summary Report. Default execution is one user and one loop so the student can demonstrate functional testing safely before increasing the load.

6. Execution Procedure

1. Install/open Apache JMeter 5.6.3.
2. Install Python dependencies with: py -m pip install -r requirements.txt
3. Start the backend with: py -m uvicorn app:app --reload --port 8000
4. Verify http://127.0.0.1:8000/ and http://127.0.0.1:8000/docs in a browser.
5. Open Taskflow_JMeter_Test_Plan.jmx in JMeter.
6. Run the test using the green Start button.
7. Open View Results Tree and Summary Report.
8. Capture screenshots showing successful requests and the Summary Report.
9. Fill the Actual Result/Status column and attach the screenshots to this report before submission.

7. Evidence to Capture

Screenshot 1 – Taskflow root endpoint in browser.
Screenshot 2 – JMeter Test Plan tree showing Thread Group and samplers.
Screenshot 3 – HTTP Request configuration for one test case.
Screenshot 4 – View Results Tree showing successful response and HTTP 200/201 status.
Screenshot 5 – Summary Report showing sample count, average response time and error percentage.

8. Results and Discussion

Record the actual results from the JMeter execution below. Do not enter assumed performance values. A functional test is considered passed when the observed response code and response body satisfy the expected result.

Test ID	Observed HTTP Code	Observed Result	Pass/Fail
TC-01			
TC-02			
TC-03			
TC-04			
TC-05			
TC-06			
TC-07			
TC-08			

9. Defects / Observations

Use this section to record any failed test, unexpected HTTP status, incorrect response data, timeout, or other observation found during execution. If all tests pass, state: “No functional defects were observed during the executed test cases.”

Observation 1: _____

Observation 2: _____

10. Conclusion

Apache JMeter was configured and used to perform structured manual functional testing of the Taskflow application. The test plan covers the main task lifecycle and a negative test. The final conclusion should be based on the actual execution results recorded in this report.

11. Submission Checklist

■ Taskflow_JMeter_Test_Plan.jmx included

■ Taskflow_Test_Cases.csv included

■ Backend source included / project accessible

■ JMeter execution screenshots added

■ Actual results and Pass/Fail status filled

■ Final PDF reviewed and submitted